

Assessing plausible rates of population growth in humpback whales from life-history data

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Abstract The rate of growth of any population is a quantity of interest in conservation and management and is constrained by biological factors. In this study, recent data on life-history parameters influencing rates of population growth in humpback whales, including survival, age at first parturition and calving rate are reviewed. Monte Carlo simulations are used to compute a distribution of rates of increase (ROIs) taking into account uncertainty in biological parameter estimates. Two approaches for computing juvenile survival are proposed, which taken into account along with other life-history data, resulted in the following estimates of the rate of population growth: Approach A: mean of 7.3%/year (95% CI = 3.5–10.5%/year) and Approach B: mean of 8.6%/year (95% CI = 5.0–11.4%/year). It is proposed that the upper 99% quantile of the resulting distribution of the ROI for Approach B (11.8%/year) be established as the maximum plausible ROI for humpback whales and be used in population assessment of the species. Possible sources of positive and negative biases in the present estimates are presented and include measurement error in estimation of life-history parameters, changes in the environment within the period these

quantities are measured, density dependence or other natural factors. However, it is difficult to evaluate potential biases without additional data. The methods presented in this study can be applied to other species for which life-history parameters are available and are useful in assessing plausibility in the estimation of population growth rates from time series of abundance estimates.

Introduction

The rate of population increase is a fundamental demographic parameter used in the assessment of wildlife populations (Skalski et al. 2005). Therefore, obtaining estimates of this quantity has important implications in conservation and management. For example, the rate at which populations grow is often required as an input parameter in population assessment and viability analysis (Mace and Lande 1991; Hilborn and Mangel 1997) as well as in ecosystem modeling (Finnof and Tschirhart 2003; Mori and Butterworth 2006). Alternatively, it can be used to compute catch quotas for exploited species (Cooke 1999; Punt and Donovan 2007) or allowable removals for threatened populations (Wade 1998).

The rate of increase (ROI) of any population of animals is fundamentally constrained by biological parameters, notably survival, birth rate and age at first parturition. Values for these parameters may vary because of density dependence, changes in carrying capacity, or other natural influences, and this may result in significantly different ROIs being observed among different populations of the same species or within the same population at different time periods. However, biological limits do exist, and given a knowledge of empirical values for these independent variables, one can assess the range of population

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growth rates that are possible (and impossible) for the species concerned.

Humpback whales (*Megaptera novaeangliae*), like many other large whale species, were heavily hunted by commercial whaling. Virtually, all populations were driven to small sizes relative to their pre-exploitation abundances (Best 1993; Clapham et al. 1999; Findlay 2001). After the complete cessation of humpback whaling operations, populations started to recover providing opportunities to develop surveys with the objective of measuring and monitoring the rate at which they grow over time (e.g. Best 1993). Currently, estimates of humpback whale population growth rates are available for almost all ocean basins (Table 1). However, some of these estimates have been questioned in light of the relatively high values observed (e.g. International Whaling Commission [IWC] 2005). For this reason, methods that incorporate life-history parameters (for example, life tables, Eberhardt 1985; Barlow 1991; Skalski et al. 2005) are useful to provide insights into the plausibility of such estimates and alternative estimates of rates at which populations increase.

Previous estimates of maximum ROIs for humpback whales based on hypothetical life-history parameters were computed by Brandão et al. (2000). These authors suggested that, under optimal circumstances, rates of growth of humpback whales could be 14.8%/year ($=0.138$ instantaneous rate). However, some of the life-history values used in their calculations were outside of the range known for the species. Clapham et al. (2001) reviewed the existing literature on these parameters for humpback whales and concluded that rates of population growth in excess of 13.4%/year ($=0.126$ instantaneous rate) were biologically unrealistic. Some of the estimates of the life-history parameters used to calculate the maximum rate of growth in Brandão et al. (2000) and Clapham et al. (2001) are inconsistent with current knowledge of the biology of humpback whales, and some of the assumptions made to estimate maximum plausible growth rates are unrealistic. For example, these authors assumed equal values for calf and non-calf survival probabilities. In addition, some of the values used in their computations were point estimates and therefore did not account for uncertainty in parameter estimates. For this reason, it is important to reconsider estimates of maximum ROIs for humpback whales within biologically and statistically plausible scenarios.

Here, we expand the analyses of Brandão et al. (2000) and Clapham et al. (2001) by reviewing new data on life-history and on factors influencing population growth in humpback whales and by incorporating uncertainty in parameter estimation. We also examine recent information on estimates of growth rates in certain populations and discuss the implications of these data for humpback whales. The methods presented here can easily be

expanded to other species and are useful in assessing plausibility in the estimation of population growth rates from time series of abundance estimates.

Review of humpback whale life-history parameters

Our review of humpback whale biological parameters is based upon literature published since the reviews by Friday (2000) and Clapham et al. (2001), as well as on available unpublished data. A summary of the compiled information is provided in Table 1. Pregnancy rates assessed from whaling catches were not included in this study because pregnancy rates are not birth rates; direct data on the latter are more reliable for use in calculating rates of increase, and such information is available from a variety of studies of living whales.

Age at first parturition

Estimates of age at first parturition (AFP) are available for humpback whale populations in the North and South Pacific, and the North Atlantic Oceans. Chittleborough (1959, 1965) used ear plugs to determine the age of humpback whales killed in the Australian fishery and concluded that both males and females attain sexual maturity (ASM) at an average age of 5. However, his estimate of the rate at which ear plug laminations were deposited (two Growth Layer Groups [GLG] per year) could not be verified using annual growth patterns of baleen. Nishiwaki (1959) reached a similar conclusion to Chittleborough, supporting an ASM of about 5 years, which was also based this on the assumption of a 2-GLG accumulation rate. A recent re-examination of the ear plug data (Best 2006) suggested that Chittleborough's interpretation of a 2-GLG rate was incorrect and that the actual rate of accumulation is one GLG per year. Consequently, both Chittleborough's and Nishiwaki's estimates of ASM were off by a factor of two; in other words, the humpback whales in their samples would have been mature at average age of about 10, not 5. Assuming a gestation period of nearly 1 year, a 10-year ASM implies an AFP of 11 years.

Humpback whales are individually identifiable based on the unique pigmentation patterns on their ventral fluke surface, and therefore information on AFP is available from photo-identification data collected from free-ranging individuals. Clapham (1992) estimated that the mean age of first parturition in Gulf of Maine humpback whales born between 1979 and 1987 was 5.9 years. This estimate appears to have increased to 8.8 years for individuals born after 1987 (Robbins 2007). The earliest age at which any female from the Gulf of Maine population has been observed with her first known calf was 5 years. Gabriele

Table 1 Estimates of life-history parameters for humpback whales

Mean	SE ^a	95% CI ^a	Region	Periods	Reference
Age at first parturition					
11.00 ^b	–	–	East and West Australia	1949–1962	Chittleborough (1959, 1965)
11.00 ^b	–	–	Japan	1959	Nishiwaki (1959)
5.90^c	0.23	–	Gulf of Maine	1979–1991	Clapham (1992)
11.80	–	10.30–13.30	Southeast Alaska	1984–2004	Gabriele et al. (2007)
8.78	2.33	–	Gulf of Maine	1992–2005	Robbins (2007)
Birth rate					
0.37	–	–	Southeast Alaska	1968–2000	Straley (unpublished data)
0.36	–	0.28–0.43	Alaska	1979–1986	Baker et al. (1992)
0.42	0.07	–	Gulf of Maine	1979–1987	Calving interval = 2.35 years, Clapham and Mayo (1990)
0.42	0.02	–	Gulf of Maine	1979–1991	Calving interval = 2.38 years, Barlow and Clapham (1997)
0.44	0.18	–	California	1986–1996	Steiger and Calambokidis (2000)
0.39	–	–	Gulf of Maine	1992–2000	Calving interval = 2.56 years, Clapham et al. (2003)
Survival—non-calf					
North Atlantic					
0.951	0.010	0.929–0.969	Gulf of Maine	1979–1986	Buckland (1990)
0.960	0.008	0.944–0.976	Gulf of Maine	1979–1991	Barlow and Clapham (1997)
0.964	0.007	0.946–0.976	Gulf of Maine	1979–1995	Rosenbaum et al. (2002)
0.957	0.028	0.854–0.988	Western Greenland	1988–1993	Larsen and Hammond (2004)
0.950	0.011	0.928–0.972	Gulf of Maine	1992–2000	Clapham et al. (2003)
0.925	–	0.891–0.950	Gulf of Maine	2000–2005	Robbins (2007)
North Pacific					
0.937	–	0.921–0.95	Southeast Alaska	1979–1995	Mizroch et al. (2004)
0.952	–	0.941–0.961	Southeast Alaska/Hawaii	1979–1996	Mizroch et al. (2004)
0.957	–	0.943–0.967	Southeast Alaska/Hawaii	1979–1996	Mizroch et al. (2004)
0.953	–	0.941–0.96	Southeast Alaska/Hawaii	1979–1996	Mizroch et al. (2004)
0.951	–	0.941–0.96	Southeast Alaska/Hawaii	1979–1996	Mizroch et al. (2004)
0.984	–	0.954–0.995	Prince William Sound	1979–1996	Mizroch et al. (2004)
0.931	–	0.912–0.945	Alaska/Hawaii	1979–1996	Mizroch et al. (2004)
0.963	–	0.944–0.978	Hawaii	1980–1996	Mizroch et al. (2004)
0.963	–	0.944–0.976	Hawaii	1980–1996	Mizroch et al. (2004)
Australia					
0.966	–	0.870–1.000	Eastern Australia	1988–1996	Chaloupka et al. (1999)
Survival—calf					
0.85	–	–	Gulf of Maine	1979–1991	6- to 18-month survival, Barlow and Clapham (1997)
0.818	–	0.482–0.977	North Pacific Ocean	1979–1995	0-to 6-month survival, Gabriele et al. (2001)
0.702	–	–	Gulf of Maine	1979–1996	Whales ages 1–5 years, Rosenbaum et al. (2002)
0.664	–	0.517–0.784	Gulf of Maine	2000–2005	6- to 18-month survival, Robbins (2007)

Numbers highlighted in bold were parameter estimates used in this study for the computation of ROIs

^a “–” indicates SE or 95% CI were not available in the original source

^b Corrected by Best (2006) to account for misinterpretation in growth layer group (GLG) readings. Based on corrected estimate of maturity at 10 years plus 1 year for gestation; b SE re-computed from data provided in Clapham et al. (1992)

et al. (2007) used re-sighting histories of 10 individually identified female humpback whales from Southeast Alaska to estimate AFP. The females concerned were sighted with their first observed calf at a mean of 11.8 years (range: 8–16 years), which was significantly older than those observed by Clapham (1992) (range: 5–7 years). However, the latter and the more recent rate of 8.8 years (range: 9–13 years) from the Gulf of Maine are more consistent with the corrected version of Chittleborough's (1959, 1965) estimate.

Birth rates

From calculations of non-calf female survival rate and birth interval probabilities, Clapham et al. (2003) estimated an average inter-birth interval of 2.56 years for Gulf of Maine whales between 1992 and 2000; this corresponds to a birth rate of 0.39 (1/2.56) calves (male and female) per mature female per year. This rate was not significantly different from the estimate of 0.42 calves/year (calving interval = 2.38) calculated with the same method for the same population over the period 1979–91 (Barlow and Clapham 1997).

Three birth rate estimates for Alaska and California humpback whales ranged from 0.37 to 0.44 calves per mature female per year (Baker et al. 1992; Steiger and Calambokidis 2000; Straley in prep.); all were derived from photo-id mark-recapture data. The highest observed rate was 0.44 from humpback whales in California for the time period 1986–1996 (Steiger and Calambokidis 2000).

Female humpback whales most commonly give birth 2 years after a previous parturition but are known to be capable of calving at an interval of only 1 year (i.e. being simultaneously lactating and pregnant). Various rates for 1-year calving have been reported from longitudinal studies: Clapham and Mayo (1990) reported two one-year intervals out of 55 intervals (6.3%) recorded in the Gulf of Maine between 1979 and 1987. Robbins (2007) reported that consecutive calving made up less than 2% of calving events observed in the Gulf of Maine between 1989 and 2005. Glockner-Ferrari and Ferrari (1990) reported a higher rate of consecutive calving (7 of 50 intervals, or 14%) from the Hawaiian breeding grounds. However, their study was based upon an almost exclusive focus on mothers and calves in their sampling, which would inevitably bias results toward shorter intervals.

Survival

Survival rates in baleen whales are often reported as calf and non-calf survival rates (e.g. Bradford et al. 2006). The former generally corresponds to the survival in the first

year, whereas the latter refers to the survival from age 1 onwards.

Calf survival

Calf survival can be divided into two phases: (1) from birth to arrival to the feeding grounds (also known as “between breeding grounds and summer feeding areas”) and (2) after arrival on feeding grounds to the arrival in the breeding grounds in the following year. We refer to these as 0- to 6-month survival and 6- to 12-month calf survival, respectively. In general, humpback whale calves do not become consistently identifiable until ~6 months of age when they arrive on the feeding grounds. Therefore, it is not possible to conduct photo-identification studies on calves less than 6 months old unless identification of mothers is used as a surrogate to identify calves. This method was used by Gabriele et al. (2001) to infer the mortality rate of humpback whale calves in their first 6 months by comparing photo-identification data between Hawaii (the wintering ground) and Alaska (the feeding ground to which calves go with their mother in the former's natal year). Mothers observed in Hawaii during the winter were resighted with or without their calves in Alaskan waters the following summer or fall. The minimal and most precise estimate was 0.15 (95% CI: 0.032–0.378); this excluded all sightings made in the fall (when calves may have already separated from their mothers). The maximal rate (using all sightings) was 0.241 (95% CI: 0.103–0.434). The authors suggested that an intermediate rate, which excluded fall sightings as well as single sightings of individual females, was likely to be the most realistic. This resulted in a mortality estimate of 0.182 (95% CI: 0.023–0.518; survival rate = 0.818) in the first 6 months of age.

Rosenbaum et al. (2002) provided an estimate of Gulf of Maine humpback whale “calf survival” of 0.702, but this actually corresponds to calves plus sexually immature individuals (ages 1–5 years). More recently, Robbins (2007) provided a mean estimate of 0.664 (95% CI: 0.517–0.784) for calves in this same population. Because this study was conducted in the feeding grounds, this estimate actually corresponds to individuals of 6–18 months in age.

Non-calf survival

Multiple estimates of non-calf survival are available for humpback whales in different ocean basins (Table 1). Estimates range from 0.95 to 0.985, but all confidence intervals overlap to a great extent, indicating that they are not statistically different from each other. In many cases (e.g. the Gulf of Maine), multiple estimates correspond to the same population, but the datasets used to compute

survival probabilities differ in the temporal and geographic span of the dataset. Estimates of survival were usually derived from feeding grounds, and therefore they incorporate the survival of calves post six months, since once calves successfully reach high latitudes, they are individually identifiable and thus are available for resighting in subsequent years. As such, estimates of adult survival from feeding grounds represent a combination of calf and non-calf survival rates. The only feeding ground-derived estimate that specifically breaks out the survival of calves are those by Clapham et al. (2003) and Robbins (2007) for the Gulf of Maine.

Computing growth rates

We used the humpback whale life-history data to compute estimates of growth rate with the following model:

$$e^{p \cdot r} = e^{(t_p - 1) \cdot r} \cdot S + pq_f \cdot S_c \cdot S^{(t_p - 1)},$$

where t_p —age at first parturition S —non-calf survival rate S_c —calf survival rate p —birth rate q_f —sex ratio at birth r —rate of increase.

The model formulation is the one used in the computations of Brandão et al. (2000) and Clapham et al. (2001) and provides results that are equivalent to the parameterization of the Lotka equation proposed by Skalski et al. (2008) to compute growth rates of mammal populations with a stable age distribution. An advantage of the model described earlier is that it does not require estimates of longevity for long-lived animals (such as large baleen whales), a parameter needed in the typical Lotka equation (e.g. Eberhardt 1985). In this study, growth rates were estimated using Monte Carlo methods (e.g. Barlow 1991). Life-history parameters were drawn from a sampling distribution (Table 2) and solved for the value of r . This process was repeated 10,000 times. For some parameters, bounds were imposed on the sampling distributions to prevent draws of unrealistic values (Table 2). Rates of increase can be expressed as an instantaneous rate (r , such as in the above-mentioned model), or as a finite rate (λ), where $\lambda = e^r$. In this study, ROI is expressed as the

percentage increment in population abundance from 1 year to another and is computed as: $\text{ROI} = (\lambda - 1) \cdot 100$.

Sampling distributions for the life-history parameters were determined from the literature review described earlier (Table 1). Selecting appropriate estimates to use in a simulation is a difficult task. Initially, one would consider combining all existing life-history parameters (e.g. by averaging across all estimates) so that all the available information can be incorporated in the analysis. However, this poses some potential issues for the kind of analysis intended here. First, for each life-history parameter, it would result in the combination of estimates observed from populations reduced to different depletion levels and with different recovery histories (e.g. North Atlantic and North Pacific populations). Second, it would combine estimates of biological parameters obtained of the same population at different periods in their recovery history. For example, the estimate of age at first parturition was different earlier (1979–1991, Clapham 1992) than later (1992–2005, Robbins 2007) in the Gulf of Maine. During the period considered, this population may have increased by at least 20% (Stevick et al. 2003; Punt et al. 2006). Because the main objective of this paper was to investigate maximum possible rates of increase, the following criteria were used in selecting life-history parameters for the simulations:

- (1) If multiple parameter estimates were available for the same population, earlier rather than later estimates were preferred because these were likely obtained when the population was closer to its maximum depletion levels and therefore life-history parameters are likely less influenced by compensatory density-dependence mechanisms.
- (2) We tended to select estimates with more extreme values, regardless of population status.

With that in mind, the following life-history parameters were selected:

Age of first parturition (t_p = AFP)

The estimate of Clapham (1992, [AFP = 5.9 years, SE = 0.23 years]) was selected as the most appropriate estimate

Table 2 Sampling distribution and parameter values of humpback whale life-history parameters used in the Monte Carlo simulations

Parameter	Sampling distribution	Mean	SE ^a	Bounds
Non-calf survival	Beta	0.984	0.009	Upper: <1
Calf survival, approach A	Beta	0.670	0.14	Upper: <1
Calf survival, approach B	Beta	0.811	0.12	Upper: <1
Age at first parturition	Normal	5.90	0.23	Lower: 5 years
Birth rate	Normal	0.42	0.02	Upper: 0.5

^a Standard error (SE) of the estimate of calf and non-calf survival was re-computed using the methods and data presented in Gabriele et al. (2001, p. 592). Non-calf survival SE was provided by S. Mizroch (in litt, 21 December 2009)

for this study as it covers the earliest period in which direct observations (photo-identification) were made and is the lowest value of AFP available for any humpback whale population. Estimates of AFP from Chittleborough (1959, 1965) and Nishiwaki (1959) were obtained while populations were declining due to whaling, but before they reached their maximum depletion levels (Johnston and Butterworth 2005; Leaper et al. 2010). Therefore, they likely do not reflect AFP at the lowest observed population levels. The remaining estimates from photo-identification data (Gabriele et al. 2007; Robbins 2007) were obtained relatively recently, likely after some (presumed) greater population recovery.

The sampling distribution of AFP was left-truncated at 5 years (a lower bound) because this is the lowest AFP ever observed for any individual humpback whale (Clapham and Mayo 1990; Clapham 1992; Robbins 2007).

Annual birth rate (p)

The birth rate was obtained from the reciprocal of the calving interval of Barlow and Clapham (1997) (2.38 years, SE = 0.10 years). The resulting value (0.42 calves/year, SE = 0.02 calves/year) is considered the preferred estimate because it uses an unbiased estimate of reproductive rates (the birth interval approach, Barlow and Clapham 1997) and because it covers a relatively early period (1979–1991) when the population was believed to be more depleted. This figure is very similar to the Clapham and Mayo (1990) estimate of calving interval (2.35 years, SE = 0.70 years) for the same region and period but is more precise.

The birth rate estimates of Clapham et al. (2003) for the Gulf of Maine and Straley (unpublished data) for the North Pacific Ocean also used a birth interval approach, but were from a more recent period; therefore, some recovery may have taken place in these populations. Robbins (2007) calculated annual calving probabilities using multi-state modeling techniques, but her estimates are also based on samples collected at a later period. The remaining estimates (Baker et al. 1992; Steiger and Calambokidis 2000) were estimates of crude birth rate.

An upper bound (truncation) on the birth rate sampling distribution was set at 0.5, which assumes a calving interval of 2 years. While individual humpback whales have occasionally been observed to reproduce at a one-year interval, this has never been observed on a population basis. Because Clapham and Mayo (1990) indicated that most whales in Gulf of Maine produced one calf every 2 years, and because the preferred estimate of birth rate comes from this population, we set the bound at this value.

Sex ratio (q_f)

This was assumed to be 1:1, based upon data presented in Clapham et al. (1995).

Survival—non-calf (S)

A number of estimates of non-calf survival exist for humpback whales (Table 1). In general, they cover relatively long time spans (7–16 years with both North Atlantic and North Pacific data starting in 1979). Point estimates range from 0.925 to 0.984, but confidence intervals overlap to a great extent indicating that no single estimate is statistically different from the others. In the interest of exploring maximum rates of increase, we selected the highest estimate ($S = 0.984$, SE = 0.009, Mizroch et al. 2004. Mizroch, in litt 21 December 2009) as the most appropriate for our analysis.

Survival—calf (S_c)

Estimates of first year (0–12 months) annual survival are not available for humpback whales. Therefore, the following approaches were used to compute a value for this parameter:

Approach A

We squared the 0- to 6-month survival estimate (0.818, SE = 0.122) from Gabriele et al. (2001) to obtain an annual calf survival ($S_c = 0.670$, SE = 0.142¹). This is considered a conservative estimate because it assumes that the 6- to 12-month survival equals the 0- to 6-month survival. Unlike other baleen whales (e.g. Antarctic minke whale, *Balaenoptera bonaerensis*), which become independent from their mothers within a few months of age (e.g. Williamson 1975), humpback whale calves remain with their mothers for one or sometimes 2 years (Clapham and Mayo 1987). Therefore, it is expected that mortality will be highest in the first 6 months of age for the latter.

Approach B

We multiplied the existing 0- to 6-month survival from Gabriele et al. (2001) as mentioned earlier by the square root of the non-calf survival. This assumes that the 6- to 12-month survival equals the non-calf survival, and thus can be seen as a more liberal approach. This resulted in an estimate of $S_c = 0.811$ (SE = 0.12²).

¹ SEs for calf survival in Approaches A and B were computed using the delta method assuming the 0–6 month and the 6–12 month survival probabilities to be independent.

Calf survival was constrained to be always less than adult survival in the simulations.

Simulation results

Simulated life-history parameters and ROIs for humpback whales are reported in Table 3. Simulations resulted in mean estimates of ROI of 7.3%/year (95% CI = 3.5–10.5%/year) for approach A and mean of 8.6%/year (95% CI = 5.0–11.4%/year) for approach B. The 99% percentile (11.8%/year) of the ROI distribution obtained for Approach B (the one with a less conservative estimate of calf survival) is proposed here as a maximum (or an upper bound) ROI for humpback whales.

Discussion

ROIs and life-history parameters

In this study, we considered what we believe are the most optimistic life-history parameters to compute a distribution of plausible growth rates for humpback whale as a species. The mean ROIs computed here (mean of 7.3%/year and 8.6%/year) are broadly similar to growth rate estimates in all ocean basins (Table 4), suggesting that most humpback whale populations were growing near their maximum potential when the first population monitoring surveys started. However, there are a few exceptions. The two most recent estimates from the North Atlantic (4%/year, Gulf of Maine, Clapham et al. (2003) and 3.1%/year, North Atlantic, Stevick et al. (2003) show growth at a lower rate. This is consistent with the most recent estimates of life-history parameters computed for this population, in particular an increase in AFP (Robbins 2007) and in calving interval (Clapham et al. 2003). Exceptions also include population growth rates that are at or above (sometimes well above) the distributions of simulated ROI values and

the maximum ROI (11.8%/year) proposed here. These include point estimates and confidence intervals obtained for Iceland (Sigurjonsson and Gunnlaugsson 1990) and for the Antarctic (Matsuoka et al. 2004; Table 4). Our analysis strongly suggests that the point estimates of these rates are not plausible as measures of true population growth. The reasons these rates are inflated may include undetected survey bias, population sub-structuring, immigration from other populations or other factors, and they deserve further investigation.

Maximum rates of increase for humpback whales were proposed in earlier reviews by Brandão et al. (2000) (14.8%/year) and Clapham et al. (2001) (13.4%/year). These estimates were based upon a 0.99 annual survival probability for calves and non-calves, a successful pregnancy rate of 0.50 and an AFP of 5 years. Most of these parameter values are too optimistic and perhaps unrealistic. Average AFPs of 5 years have never been recorded for any population of humpback whales. A mean AFP of 5.9 years was observed in the Gulf of Maine population from 1979 to 1991, but in recent years, this quantity has increased to 8.8 years. These changes may be reflected in the observed decrease in population growth rate from 6.5% in 1979–1991 to a best estimate of 4% in a later (1992–2000) period (Clapham et al. 2003).

Population-level birth rates of 0.5 or greater have also not been observed for humpback whales. Figures presented in the literature range from 0.37 (1 successful calf every 2.7 years) to 0.44 (one successful calf every 2.3 years). While values higher than the one selected in this study (birth rate = 0.42, calving interval = 2.38) exist, they were crude birth rates and therefore may be potentially biased (Barlow and Clapham 1997).

Adult survival probabilities near 0.99 are plausible for large whales but have been reported only for very long-lived species such as the bowhead whale (*Balaena mysticetus*, 0.984 [95% CI: 0.948–1.000], Zeh et al. 2002) and the southern right whale (*Eubalaena australis*, Argentina: 0.981 [95% CI: 0.971–0.991], Cooke et al. 2001; South

Table 3 Summary of simulated humpback whale life-history parameters and rates of increase

	Non-calf survival	Calf survival	Age at first parturition	Birth rate	Rate of increase
Approach A					
Mean	0.984	0.669	5.90	0.41	7.3%/year
Median	0.985	0.680	5.90	0.42	7.5%/year
95% CI	0.963–0.996	0.366–0.905	5.43–6.35	0.33–0.49	3.5–10.5%/year
Approach B					
Mean	0.984	0.808	5.90	0.41	8.6%/year
Median	0.986	0.830	5.90	0.42	8.8%/year
95% CI	0.970–0.997	0.517–0.970	5.43–6.35	0.32–0.49	5.0–11.4%/year
Proposed maximum ROI (upper 99th percentile)					11.8%/year

Table 4 Selected estimates of rates of increase (ROI) for humpback whales worldwide

Region	Period	ROI (%/year)	95% CI	Reference
North Pacific				
North Pacific Ocean	1966–2005	6.8	–	Calambokidis et al. (2008)
Western North Pacific	1980–1996	10.0	3–16	Mizroch et al. (2004)
Northern Gulf of Alaska	1987–2003	6.6	5.2–8.6	Zerbini et al. (2006)
West coast of the USA	1990–2008	8	–	Calambokidis (2009)
North Atlantic				
Iceland	1968–1988	14.8	12.3–17.2	Sigurjonsson and Gunnlaugsson (1990)
Gulf of Maine	1979–1991	6.5	4.1–8.8	Barlow and Clapham (1997)
North Atlantic Ocean	1979–1993	3.1	2.1–4.2	Stevick et al. (2004)
Gulf of Maine	1992–2000	4.0	–	Clapham et al. (2003)
Southern Hemisphere				
Eastern Indian Ocean/W Australia	1977–1991	10.1	5.5–14.7	Bannister and Hedley (2001)
Antarctic (Circumpolar)	1978–2004	9.6	5.8–13.4	Branch (2010)
Western Pacific Ocean/E Australia	1984–2007	10.9	10.5–11.4	Noad et al. (2008)
Western Pacific Ocean/E Australia	1987–1999	10.9	10.2–11.6	Paterson et al. (2001)
IWC Management Area IV in the Antarctic	1989–2003	19.8	11.6–28.0	Matsuoka et al. (2004)
IWC Management Area V in the Antarctic	1989–2003	13.0	7.5–18.4	Matsuoka et al. (2004)
IWC Management Areas IV–V in the Antarctic	1989–2003	14.7	8.6–20.7	Matsuoka et al. (2004)
Western Indian Ocean	1991–2003	7.9	–	Findlay et al. (2004)
Western South Atlantic	1995–1998	7.4	0.6–14.5	Ward et al. (2010)
IWC Management Areas III+VI in the Antarctic	1995–2003	25.0	18.5–31.5	Matsuoka et al. (2004)
Western South Atlantic	1996–2000	30.6	2.6–60	Freitas et al. (2004)

Africa: 0.99 [95% CI: 0.983–0.997], Best et al. 2005). For other large whale species for which estimates are available, the reported numbers are lower. For example, survival of non-calf western gray whales (*Eschrichtius robustus*) has been estimated at 0.951 (95% CI: 0.917–0.972; Bradford et al. 2006). Point estimates of non-calf survival probability estimates for humpback whales have consistently fallen within the range of 0.95–0.984, in studies comprising different time periods and populations in three ocean basins (see Table 1 for details). The survival probability selected in this study was the highest figure ever observed for any humpback whale population and is similar to estimates provided for longer lived balaenid species, suggesting that greater values are very unlikely for humpback whales.

Perhaps, the most important aspect of this study was to consider a more realistic estimate of calf survival. Previous estimates of maximum growth rates for humpback whales have assumed equal survival probabilities of “juveniles” (calves) and “adults” (non-calves) (Brandão et al. 2000; Clapham et al. 2001). It is well known that mammals generally have a pattern of lower early survival relative to adult survival (Caughley 1966), which is consistent with observed survival estimates for other mammals (Kraus et al. 2005; Baker and Thompson 2006), including small cetaceans and large whales. For example,

survival estimates of free-ranging bottlenose dolphins have shown that the ratio calf/non-calf survival is nearly 0.84 ($S_c = 0.8$ –0.81, $S = 0.96$; Wells and Scott 1990). For baleen whales, this ratio seems to vary. A ratio of 0.74 has been observed for right whales ($S_j = 0.734$, $S = 0.99$; Best et al. 2005) and for gray whales ($S_c = 0.701$, $S = 0.951$; Bradford et al. 2006). In humpback whales, mean estimates of calf survival in the Gulf of Maine correspond to about 70–90% of the adult survival probability (see Table 1, Barlow and Clapham 1997; Robbins 2007). However, the humpback and gray whale estimates were obtained from feeding ground studies and actually correspond to the 6- to 18-month survival (Robbins 2007) or 6- to 18-month (Barlow and Clapham 1997; Bradford et al. 2006) survival, so the actual calf/non-calf survival ratios are actually lower. In this study, the two approaches used to select calf survival resulted in point estimates that led to calf/non-calf ratios ranging from 0.7 to 0.82, which are consistent with other cetacean populations.

Maximum plausible ROIs for humpback whales

The purpose of this paper was to assess plausible estimates of ROI given known life-history parameters and to propose

a maximum ROI taking into account existing life-history data, realistic estimates of calf survival and uncertainty in the estimates of life-history parameters. Maximum ROIs proposed by Brandão et al. (2000) (14.8%/year) and Clapham et al. (2001) (13.4%/year) were based upon extreme parameter values and unrealistic assumptions (e.g. the same values for calf and adult survival). The simulations conducted earlier provided a distribution of ROIs given the range in which parameter values could potentially be observed (e.g. their sampling distribution). In 10,000 replicates, there was not a single case in which a combination of parameters produced a ROI as high as those reported by Brandão et al. (2000) and Clapham et al. (2001).

The maximum rate at which a population can grow is of considerable interest in conservation and management. The choice of such a value is inevitably difficult, and should be established taking into consideration reliable knowledge of the biological limitations of the population. The simulations described earlier resulted in ROIs broadly consistent with existing empirical estimates of ROI obtained from analysis of mark-recapture and sighting surveys of depleted humpback whale populations. Given available life-history estimates, the proposed upper bound maximum ROI of 11.8%/year is a suitable number as it reflects a near-maximum value (i.e. there is only a 1% probability of a higher value in the simulations) and is not dependent on sample size (i.e. on the number of simulation draws). We therefore suggest this can serve as an upper bound for any empirical estimate of ROI, and we emphasize that such a high figure can be observed only with extreme and very optimistic life-history parameters. We also recommend that the range of realistic values for population increase summarized here be used to ground-truth estimates derived from recent and future survey data and to exclude (e.g. in population assessments of humpback whales) any calculated rates that fall above this frame of reference. Finally, the distribution of ROIs provided in this study may be useful as a default rate in the absence of other data on maximum growth rates. For example, current Bayesian population assessment models for humpback whales include the intrinsic rate of population growth as a parameter (e.g. IWC 2007, 2008; Jackson et al. 2008, 2010; Zerbini et al. 2010). This parameter corresponds to a theoretical maximum rate of increase a population can reach under optimal circumstances (e.g. in the absence of density dependence and when growth is not limited by the environment, see Skalski et al. 2008). A distribution of ROIs computed from life-history data using Monte Carlo methods can be used as a prior for the intrinsic growth rate in Bayesian models. Such an approach has been used to assess the status of other large whales [e.g. Antarctic Blue whales (Branch et al. 2004) and western gray whales (Bradford 2003)].

How realistic is the proposed ‘maximum’ rate?

The ROIs reported earlier, including the proposed maximum value, assume that all populations of humpback whales have similar life-history characteristics and that parameter values were obtained when populations were growing at their maximum. However, it is possible that under certain circumstances, the distributions of ROIs reported earlier are conservative and proposed maximum ROI may not reflect the actual maximum. For example, some of the biological parameters used as input data were obtained from populations that were already recovering from exploitation and therefore may have not been growing at their maximum rate due to density-dependent mechanisms. In addition, demographic parameters may fluctuate according to environmental or ecological processes, and it is possible that even if populations were growing at or near their maximum, variations in the parameters observed here were in response to the whales’ environment, and the maximum value was not observed. Negative bias may exist in life-history parameter estimation due to difficulties in sampling or unaccounted anthropogenic mortality. For example, the AFP and the birth rates used in the present analysis were derived from the feeding grounds and therefore incorporate a calf mortality component. Therefore, if measured in the wintering (breeding) grounds, AFP could be lower and birth rates could be higher than values used in this analysis. Also, both calf and non-calf survival estimates incorporate some unobserved human-related mortality such as bycatch or vessel collisions. Therefore, they may not correspond to the intrinsic survival capacity in the population, but to a number lower than that.

While it is possible that some life-history parameters values are not measured near their maximum, the simulations reported here can potentially be considered optimistic. We used only two values for survival, a calf and a non-calf rate, and attributed the latter to the component of the population older than 12 months. However, it has been shown that sexually immature animals have lower survival than mature individuals in other mammal species (Kraus et al. 2005; Baker and Thompson 2006), including humpback whales (Rosenbaum et al. 2002; Robbins 2007). Also, the simulation assumed that AFP was knife-edge; that is 100% of the individuals reproduced once they attained sexual maturity. This is also unlikely as not all individuals become sexually mature at the same age and not all mature individuals reproduce (Best et al. 2001; Cooke et al. 2001). If they did, costs of reproduction could potentially reduce survival, as suggested for the Gulf of Maine humpback whale population (Robbins 2007). Therefore, if all these factors were accounted for in the simulation presented here, it is possible that the resulting ROI values would be even lower than we have estimated. Additionally, it bears repeating that the point

estimate of our ROI distributions has a much lower ROI than the proposed maximum value. The latter corresponds to a maximum bound incorporating uncertainty, which would represent a combination of the more extreme values of the life-history parameters than have been estimated for humpback whales in any given ocean basin.

The simulations presented earlier mixed vital rates from different populations in various ocean basins. Different basins differ in productivity and other aspects that could affect population growth. In addition, there are conceivably trade-offs in life-history strategies where populations in one basin survive better but reproduce more slowly or vice versa. For this reason, one would ideally prefer to obtain maximum estimates of ROI that are population-specific. This is impractical here because estimates of some life-history parameters are not available for many populations, notably those in the Southern Hemisphere. In this respect, it is noteworthy that the ROIs computed in this paper have been based on life-history parameters of Northern Hemisphere populations.

At present, existing information is insufficient to evaluate whether and how sources of potential positive and negative bias would affect estimates of the maximum ROI for humpback whales. For this reason, we think that the values we have proposed earlier are the most realistic estimates given the available data and the assumptions being made. This is justified by the fact that, despite extensive long-term studies in many locations, there is currently no evidence that humpback whale life-history parameters can be more extreme than the values implemented in the present simulation (for example, humpback whales conceiving at less than 4 years of age or population birth rates greater than 0.50 have never been observed). Further analysis, with data collected from other populations and for longer periods of time, could provide more insights into the question of the maximum growth rate of humpback whales. Reviews of the analysis presented in this study are therefore encouraged once such data become available.

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